

Document 2: Technology and Modernization at SCCL

Introduction

Singareni Collieries Company Limited has continuously modernized its mining operations since 1889. This document outlines the technological evolution and current equipment used in SCCL's coal mining operations.

Historical Technology Timeline

Year	Technology Introduced
1889	Manual coal extraction begins
1937	Introduction of coal drilling machines
1950	Shuttle cars, belt conveyors, coal plough equipment
1975	Walking draglines, power shovels, high-capacity dumpers for opencast mining
1983	Longwall mining technology for underground operations
1989	Blasting Gallery (BG) technology introduced
1994	In-pit crushing and conveyor system (German collaboration)
2002	Surface miner technology (environmentally friendly, eliminates blasting)
2019–20	Manual coal loading completely phased out

Current Equipment Inventory

Equipment Type	Quantity
Longwall Systems	1
Continuous Miners	5
Bolter Miners	2
Side Discharge Loaders (SDLs)	107
Load Haul Dumpers (LHDs)	10
Road Headers	2
Walking Draglines	Multiple
Surface Miners	Multiple
High-Capacity Dumpers	Multiple (up to 150 tonnes)

Underground Mining Technologies

Longwall Mining

- Highly mechanized method for extracting coal in long panels
- Uses powered roof supports, shearers, and armored face conveyors
- High productivity with improved safety

Continuous Mining

- Uses continuous miner machines that cut coal and load it simultaneously
- Eliminates drilling and blasting in certain seams
- Five continuous miners currently operational

Bord and Pillar Method

- Traditional method still used in some mines
- Modernized with mechanized loaders and conveyors

Opencast Mining Technologies

Surface Miners

- Cut coal directly from the seam without drilling or blasting
- Environmentally friendly with reduced dust and noise
- Selective mining capability

Dragline Operations

- Walking draglines remove overburden (soil above coal seams)
- Large-scale earth-moving capability

In-Pit Crushing and Conveying

- Crushes coal at the mining face
- Conveys directly to surface using belt systems
- Reduces truck haulage and fuel consumption

Digital Initiatives

Initiative	Description
Mine Management Software	Custom software for production tracking and planning
Minex Software	Digitized mine plans and geological modeling
Video Conferencing	Connects all mines to headquarters
E-Procurement	Online tendering and vendor management
GPS Tracking	Real-time tracking of vehicles and equipment
Safety Monitoring	Electronic monitoring of underground conditions

Safety Innovations

- Gas detection and monitoring systems
- Automatic fire suppression systems
- Refuge chambers in underground mines
- Modern ventilation systems
- Personal protective equipment standards
- Regular safety training programs

Environmental Measures

- Afforestation programs in mined-out areas
- Water treatment and recycling
- Dust suppression systems
- Land reclamation after mining
- Solar power installations at mining facilities